

## KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2111

II

As specified in the official PDF

Source-grounded

# Fundamentals of Manufacturing and Tool Engineering

system, solidification and an overview of other casting and moulding methods

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 2111 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 2111.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

**COURSE DASHBOARD**

# Official course information

**Course information**

| Item                 | Official information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programme            | Diploma Programme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Course code          | 2111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Course title         | Fundamentals of Manufacturing and Tool Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Semester             | II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Course category      | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Periods per week     | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Periods per semester | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Credits              | As specified in the official PDF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Prerequisite         | Topic Course code Course Title Semester Basic Physics 2002A Engineering Physics for Mechanical, Structural and Industrial Applications I/II Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Explain the evolution and fundamentals of Manufacturing and Tool Engineering Understand CO2 Explain sand casting and other basic casting processes along with their applications Understand Explain the basic principle of Metal forming process, Machining and Joining process along with their CO3 Understand application Identify and classify Engineering materials and Tool materials. Describe about heat treatment and CO4 Understand modern trends in Industry CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES |
| Assessment           | Use the official assessment section reproduced below                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**Modules****Topic entries**

4 official module sections detected.

15 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

## STUDY METHOD

### How to use this lesson

#### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

#### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

#### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

#### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

**LEARNING OUTCOMES****Objectives, outcomes and mapping****Course objectives**

- system, solidification and an overview of other casting and moulding methods

**Course outcomes**

| CO  | Official outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Study level           |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 3 2 - - - 2 2 - - - 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | See official syllabus |
| CO2 | 3 3 2 2 - 2 - - - - 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | See official syllabus |
| CO3 | 3 3 2 2 2 2 - - - - 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | See official syllabus |
| CO4 | 3 2 2 - 2 3 2 - - - 2 Total 12 10 6 4 4 9 4 - - - 8 Correlation Level 3 2.5 2 2 2 2.25 2 - - - 2 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total Diploma Curriculum Revision 2026 2111 Page #1 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Evolution and basics of Introduction and classification of manufacturing processes, safety | See official syllabus |

**CO-PO mapping evidence**

| Row | Extracted official mapping |
|-----|----------------------------|
|-----|----------------------------|

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3 2 - - - 2 2 - - - 2  |
| CO2 | CO2 3 3 2 2 - 2 - - - - 2  |
| CO3 | CO3 3 3 2 2 2 2 - - - - 2  |
| CO4 | CO4 3 2 2 - 2 3 2 - - - 2  |

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

| Module   | Official heading                                                                        | Topic entries | Hours        |
|----------|-----------------------------------------------------------------------------------------|---------------|--------------|
| Module 1 | manufacturing and Tool 15 0                                                             | 4             | As specified |
| Module 2 | Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 | 3             | As specified |
| Module 3 | 15 0                                                                                    | 2             | As specified |
| Module 4 | Tool Materials and Modern Trends 15 0                                                   | 6             | As specified |

**MODULE 1**

# manufacturing and Tool 15 0

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: manufacturing and Tool 15 0
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ manufacturing and Tool 15 0

manufacturing and Tool 15 0 is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify manufacturing and Tool 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, manufacturing and tool 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                       |
|---------------------|-----------------------------------------------------------------------|
| Definition/identity | What manufacturing and Tool 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application               |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-എന്നുണ്ടാകുന്ന manufacturing and Tool 15 0

എന്നതിന്റെ definition/meaning എഴുതണം. എന്നാൽ എങ്ങനെ  
എങ്ങനെ, steps, application എങ്ങനെ എഴുതണം. Technical terms  
English-ൽ എഴുതണം എങ്ങനെ എഴുതണം.

## ▼ practices in manufacturing

practices in manufacturing is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify practices in manufacturing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, practices in manufacturing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                      |
|---------------------|----------------------------------------------------------------------|
| Definition/identity | What practices in manufacturing means in the official course context |
| Study lens          | Principle → components or stages → result → application              |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-പ്രായോഗിക പ്രവർത്തനങ്ങൾ in manufacturing

പ്രായോഗിക പ്രവർത്തനങ്ങൾ definition/meaning പ്രായോഗിക പ്രവർത്തനങ്ങൾ. പ്രായോഗിക പ്രവർത്തനങ്ങൾ, steps, application പ്രായോഗിക പ്രവർത്തനങ്ങൾ പ്രായോഗിക. Technical terms English-ൽ പ്രായോഗിക പ്രവർത്തനങ്ങൾ.

## ▼ Engineering

Engineering is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Engineering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, engineering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What Engineering means in the official course context   |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-എം Engineering പദപരിചയം പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-എം പദപരിചയം പദപരിചയം.

## ▼ Basics of casting, sand casting process, pattern and moulding sand,

Basics of casting, sand casting process, pattern and moulding sand, is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Basics of casting, sand casting process, pattern and moulding sand, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, basics of casting, sand casting process, pattern and moulding sand, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



**MODULE 2**

# **Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0**

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## **Learning goals**

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## **Key facts**

- Official module title: Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## ▼ Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0

Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 is an official topic in Module 2 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, casting processes gating and risers, casting defects, special casting and moulding 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 definition/meaning . . . , steps, application . . . . Technical terms English- . . . .

## ▼ processes

processes is an official topic in Module 2 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What processes means in the official course context     |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്ന പ്രക്രിയകൾ നിർവ്വചനം/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms English-ൽ നിർവ്വചിക്കുക.

## ▼ Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press

Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press is an official topic in Module 2 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, metal forming, machining and metal forming process, machining operations, press work and press should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide



**MODULE 3****15 0**

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

**Learning goals**

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

**Key facts**

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

## ▼ Press work tool, Metal joining processes

Press work tool, Metal joining processes is an official topic in Module 3 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Press work tool, Metal joining processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, press work tool, metal joining processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                    |
|---------------------|------------------------------------------------------------------------------------|
| Definition/identity | What Press work tool, Metal joining processes means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-പ്രസ്സ് വർക്ക് ടൂൾ, മെറ്റൽ ജോയിൻഗ് പ്രോസസ്സ്. പ്രസ്സ് വർക്ക് ടൂളിന്റെ definition/meaning എഴുതുക. പ്രസ്സ് വർക്ക് ടൂളിന്റെ പ്രയോഗം, steps, application എഴുതുക. പ്രസ്സ് വർക്ക് ടൂളിന്റെ Technical terms English-ൽ എഴുതുക.

## ▼ Tool Materials and Heat treatment, Jigs Fixtures and Inspection,

Tool Materials and Heat treatment, Jigs Fixtures and Inspection, is an official topic in Module 3 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Tool Materials and Heat treatment, Jigs Fixtures and Inspection, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, tool materials and heat treatment, jigs fixtures and inspection, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|



**MODULE 4**

# Tool Materials and Modern Trends 15 0

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Tool Materials and Modern Trends 15 0
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## ▼ Tool Materials and Modern Trends 15 0

Tool Materials and Modern Trends 15 0 is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Tool Materials and Modern Trends 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, tool materials and modern trends 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| Definition/identity | What Tool Materials and Modern Trends 15 0 means in the official course context |



## ▼ Surface Finishing, Powder Metallurgy, CAD/CAM & CNC

Surface Finishing, Powder Metallurgy, CAD/CAM & CNC is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Surface Finishing, Powder Metallurgy, CAD/CAM & CNC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, surface finishing, powder metallurgy, cad/cam & cnc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Definition/identity | What Surface Finishing, Powder Metallurgy, CAD/CAM & CNC means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-പ്പ് Surface Finishing, Powder Metallurgy, CAD/CAM & CNC പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-പ് പദപരിചയം പദപരിചയം.

## ▼ Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                             |
|---------------------|-------------------------------------------------------------|
| Definition/identity | What Detailed Syllabus means in the official course context |
| Study lens          | Principle → components or stages → result → application     |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്ന Detailed Syllabus പരസ്യപ്പെടുത്തിയിരിക്കുന്നു. definition/meaning പരസ്യപ്പെടുത്തിയിരിക്കുന്നു. പരസ്യപ്പെടുത്തിയിരിക്കുന്നു, steps, application പരസ്യപ്പെടുത്തിയിരിക്കുന്നു പരസ്യപ്പെടുത്തിയിരിക്കുന്നു. Technical terms English-ൽ പരസ്യപ്പെടുത്തിയിരിക്കുന്നു.

## ▼ Mode of

Mode of is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                         |
|---------------------|---------------------------------------------------------|
| Definition/identity | What Mode of means in the official course context       |
| Study lens          | Principle → components or stages → result → application |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നിട്ടുള്ള മോഡ് ഓഫ് ഡിഫിനിഷൻ/മീനിംഗ്, ഫോർമുല, ഫോർമുല, steps, application, application, application, application. Technical terms English-ൽ എഴുതുക.

## ▼ Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                |
|---------------------|----------------------------------------------------------------|
| Definition/identity | What Mapped Instructional means in the official course context |
| Study lens          | Principle → components or stages → result → application        |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-എന്നിവിടെ Mapped Instructional ഉള്ളതിനെക്കുറിച്ച് definition/meaning ഉള്ളതുകൊണ്ട്. ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്, steps, application ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്. Technical terms English-ൽ ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്.

## ▼ Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Definition/identity | What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context |

| Aspect        | What to prepare                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Study lens    | Principle → components or stages → result → application                           |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-Subtopic Student Learning Outcome delivery Taxonomy Level definition/meaning. Steps, application. Technical terms English-.

### ► Official syllabus detail and terminology (expand in digital version)

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**RAPID REFERENCE****Concept and formula bank****Answer structure**

Definition → Principle →  
Components/steps →  
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

**Practical record**

Aim → Apparatus → Procedure  
→ Observation → Result →  
Safety

Use this sequence for laboratory, workshop and field activities.

**RAPID REVISION****Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 2 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 3 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 4 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**OFFICIAL SELF-LEARNING**

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Study the evolution of manufacturing (Craft → Mass → Automation). Prepare a comparative chart showing features,
- advantages and limitations of each stage. Refer industrial examples from automotive and aerospace sectors.
- Prepare a classification chart of manufacturing processes (Primary, Secondary, Finishing). Collect real component
- samples/images and identify the manufacturing process used.
- Study safety practices in manufacturing and tool rooms. Prepare a safety checklist for a tool room and identify potential
- hazards with preventive measures.
- Prepare short notes on basic manufacturing terminology and industrial applications. Visit a nearby workshop/tool room (if
- possible) or watch industrial videos and document observations. Module II
- Study the sand casting process in detail. Draw and label a sand mould layout including cope, drag, core, gating and riser. 6
- Prepare a comparison table of different pattern types and pattern materials. Solve numerical problems related to pattern
- allowances (conceptual understanding).
- Study moulding sand composition and properties. Prepare a chart linking properties (permeability, refractoriness, strength)
- with casting quality.
- Analyze common casting defects with causes and remedies. Prepare a defect identification chart with sketches/images.
- Study other casting processes (Die casting, Centrifugal casting, Injection moulding, Blow moulding) and compare with sand 6
- casting. Module III
- Study metal forming processes (Hot vs Cold working). Prepare a comparison chart and list real industrial applications. 6

- Study machining processes (Turning, Milling, Drilling). Draw tool geometry diagrams and identify tool angles. Watch CNC

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

|               |               |             |               |               |            |     |
|---------------|---------------|-------------|---------------|---------------|------------|-----|
| – Theory      |               |             |               |               |            |     |
| CA3           | CA4           | CA5         | CA1           | CA6           | CA2        | ESE |
| Mode          | Attendance    |             | Self-learning | Self-learning |            |     |
| Self-learning | Self-learning |             | Series        | Series        |            |     |
| Written       |               |             |               |               |            |     |
| activities    | activities    |             | activities    |               | activities |     |
|               |               | Exam        | Exam          |               | Exam       |     |
| (Module 3)    | (Module 4)    | (Module 1)  | (Module 2)    |               |            |     |
| (theory)      |               | (2 modules) | (2 modules)   |               |            |     |
| Duration      |               |             |               |               |            |     |
| 2 hours       | 2 hours       | 2.5 hours   |               |               |            |     |
| Exam mark     |               |             | 15            |               | 15         |     |
| 15            | 15            | 50          |               | 50            |            | 60  |
| Converted to  |               |             |               |               |            |     |
| 20            | 20            |             |               |               |            |     |
| Marks         |               | 5           |               |               |            |     |
| 15            |               |             |               | 20            |            |     |
| 60            |               |             |               |               |            |     |
| Total         |               |             |               |               |            |     |
| 40            |               |             |               |               |            | 60  |
| Tentative     | End of        |             |               |               |            |     |
| End of        |               |             |               |               |            |     |
|               |               | By 4th week |               | By 7th week   |            |     |
| By 11th week  | By 14th week  | 8th week    |               | 15th week     |            |     |

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

**ACTIVE RECALL**

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

- ▶ **Define or explain manufacturing and Tool 15 0. (3 marks)**
- ▶ **Define or explain practices in manufacturing. (3 marks)**
- ▶ **Define or explain Engineering. (3 marks)**
- ▶ **Define or explain Basics of casting, sand casting process, pattern and moulding sand,. (3 marks)**

### Module 2

- ▶ **Define or explain Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0. (3 marks)**
- ▶ **Define or explain processes. (3 marks)**
- ▶ **Define or explain Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press. (3 marks)**

### Module 3

- ▶ **Define or explain Press work tool, Metal joining processes. (3 marks)**

► **Define or explain Tool Materials and Heat treatment, Jigs Fixtures and Inspection,. (3 marks)**

## **Module 4**

► **Define or explain Tool Materials and Modern Trends 15 0. (3 marks)**

► **Define or explain Surface Finishing, Powder Metallurgy, CAD/CAM &CNC. (3 marks)**

► **Define or explain Detailed Syllabus. (3 marks)**

► **Define or explain Mode of. (3 marks)**

**PRACTICE ONLY**

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **manufacturing and Tool 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Press work tool, Metal joining processes**.

3-mark explanation: Explain one principle, process or comparison from this module.

### Module 4

1-mark definition: State the meaning of **Tool Materials and Modern Trends 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

**REFERENCE VOCABULARY**

# Glossary

**Study glossary**

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

**SOURCE-SUPPORTED REFERENCES**

## References and web resources

**References listed in the official PDF**

References are included in the official source PDF.

**Web resources listed in the official PDF**

- URL Modules Covered
- IIT Madras Manufacturing Process Lectures (YouTube/NPTEL) Casting, Forming, Machining
- MIT OpenCourseWare - <https://ocw.mit.edu> Manufacturing & Materials Tool Engineering & Modern SME - <https://www.sme.org> Manufacturing
- Autodesk Learning Resources - <https://www.autodesk.com> CAD/CAM & CNC

**IN-PAGE SEARCH****Search this lesson**

Prepared from the supplied official SITTTR syllabus PDF for Course 2111. Verify institutional updates against the current official curriculum.